

SHAOGUAN, China

WMO: 590820

Lat: 24.6703N

Lon: 113.6082E

Elev: 122

StdP: 99.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.3	(c) 3.5	(d) -5.5	(e) 2.4	(f) 9.4	(g) -3.0	(h) 3.0	(i) 9.2	(j) 7.4	(k) 9.2	(l) 6.8	(m) 9.8	(n) 2.0	(o) 0	(p) 0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.2	(c) 35.3	(d) 25.9	(e) 34.4	(f) 25.8	(g) 33.5	(h) 25.6	(i) 27.4	(j) 32.3	(k) 27.0	(l) 31.7	(m) 26.6	(n) 31.1	(o) 3.0	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.9	29.2	25.8	21.4	28.9	25.5	21.0	28.7	87.9	32.8	86.0	31.9	84.4	31.2	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.2	(c) 5.3	(e) 0.0	(f) 37.3	(g) 1.4	(h) 0.9	(i) -1.0	(j) 37.9	(k) -1.8	(l) 38.4	(m) -2.6	(n) 39.0	(o) -3.7	(p) 39.6		
(a) 7.1	(b) 6.2	(c) 5.3	(e) -0.8	(f) 27.9	(g) 1.5	(h) 0.7	(i) -1.8	(j) 28.4	(k) -2.7	(l) 28.8	(m) -3.5	(n) 29.2	(o) -4.6	(p) 29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	10.7	13.0	16.1	21.3	25.1	27.7	29.3	29.0	26.8	23.0	17.6	12.2
	DBStd	7.31	3.96	5.08	4.34	3.65	2.64	2.08	1.69	1.72	2.49	2.92	3.65	3.44
	HDD10.0	81	38	23	4	0	0	0	0	0	0	0	1	15
	HDD18.3	759	237	161	97	16	0	0	0	0	0	3	54	192
	CDD10.0	4106	61	108	193	339	467	530	598	588	504	404	230	84
	CDD18.3	1744	2	13	28	105	209	280	339	329	254	148	33	3
	CDH23.3	16809	4	57	112	597	1674	2900	4043	3761	2431	1030	189	10
CDH26.7	6539	0	4	9	116	526	1096	1841	1656	956	305	31	0	
Wind	WSAvg	2.1	2.1	2.2	2.1	2.2	2.1	2.4	2.4	2.0	1.9	2.0	2.0	2.1
Precipitation	PrecAvg	1713	80	89	177	211	294	290	160	145	94	47	67	58
	PrecMax	2433	318	228	475	378	833	529	481	332	192	197	238	234
	PrecMin	1163	3	4	72	38	81	137	35	32	10	0	1	2
	PrecStd	312	67	57	81	92	155	124	106	76	50	56	62	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	27.4	27.7	31.0	33.6	35.0	36.7	36.6	35.4	33.2	29.6	24.8
		MCWB	17.7	20.0	20.6	23.3	25.0	26.3	26.0	25.8	25.8	24.2	21.5	19.3
	2%	DB	21.4	25.2	26.1	29.3	32.2	33.8	35.4	35.3	34.0	31.1	27.4	22.3
		MCWB	16.3	19.0	20.0	22.5	24.5	25.7	25.9	25.9	25.3	22.6	20.1	16.6
	5%	DB	19.4	23.2	24.5	28.0	31.0	32.7	34.4	34.3	32.8	29.8	25.7	20.0
		MCWB	15.0	18.2	19.5	22.1	24.1	25.5	26.0	25.8	25.1	21.8	18.8	14.7
	10%	DB	17.2	21.1	23.0	26.7	29.6	31.6	33.3	33.3	31.6	28.4	23.7	18.0
		MCWB	13.3	17.1	18.8	21.7	23.9	25.5	25.9	25.7	24.7	21.1	17.9	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	21.6	23.0	24.9	26.9	27.6	27.8	27.9	27.2	25.5	23.2	21.1
		MCDB	21.9	25.5	25.8	28.8	31.3	32.1	33.1	33.3	32.5	30.7	27.3	23.3
	2%	WB	17.5	20.0	21.6	24.0	26.0	27.0	27.3	27.3	26.6	24.6	22.0	18.1
		MCDB	20.4	23.8	24.5	27.7	29.9	31.2	32.4	32.3	31.4	28.6	25.4	20.1
	5%	WB	15.8	18.7	20.5	23.3	25.5	26.6	26.9	26.8	26.1	23.8	20.9	16.2
		MCDB	18.9	22.4	23.3	26.3	29.0	30.6	31.8	31.6	30.5	27.4	23.8	19.0
	10%	WB	14.0	17.4	19.4	22.7	24.9	26.2	26.6	26.4	25.6	23.1	19.5	14.3
		MCDB	16.5	20.8	22.2	25.5	28.1	30.0	31.1	30.8	29.5	26.3	22.3	17.2
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.7	6.2	6.2	6.5	6.2	7.2	7.6	7.8	8.6	8.5	8.2
		MCDBR	11.3	10.5	9.7	8.8	9.0	8.1	9.0	9.5	9.7	10.6	11.7	11.8
		MCWBR	5.9	4.6	4.3	3.4	2.9	2.2	2.2	2.3	2.5	3.2	4.5	5.8
	5% WB	MCDBR	8.9	8.5	7.1	6.4	6.6	6.6	7.8	8.1	8.5	7.9	8.2	8.4
		MCWBR	5.3	4.6	4.4	3.4	2.8	2.5	2.5	2.6	2.5	2.9	4.1	4.9
Clear-Sky Solar Irradiance	taub	0.629	0.685	0.796	0.778	0.642	0.606	0.578	0.634	0.654	0.671	0.587	0.580	
	taud	1.643	1.583	1.429	1.497	1.784	1.876	1.945	1.789	1.724	1.637	1.778	1.757	
	Ebn at Noon	620	624	584	612	701	721	742	701	672	625	652	640	
	Edh at Noon	227	257	314	299	224	203	189	221	230	240	196	196	
All-Sky Solar Radiation	RadAvg	2.38	2.43	2.59	2.94	3.63	3.96	4.98	4.75	4.28	3.85	3.03	2.72	
	RadStd	0.48	0.52	0.44	0.53	0.56	0.56	0.49	0.36	0.33	0.52	0.48	0.55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	+0.35	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A	N/A	+55

Nomenclature: See separate page